SOMAIYA
VIDYAVIHAR

K J Somaiya Institute of Technology
An Autonomous Institute Permanently Affiliated to the University of Mumbai

Srushti Milap Patil (BS4 (66))

EXPERIMENT NO. 01

Program :

```
#include <stdio.h>

int stack[100], top = -1;

void Push(int data)
{
    if (top == 99)
    {
        printf("Stack overflow\n");
    }
    else
    {
        stack[++top] = data;
        if (data >= 45)
            printf("Extreme Red Heatwave Alert\n");
        else if (data >= 40)
            printf("Orange Heatwave Alert\n");
        else if (data >= 35)
            printf("Yellow Heatwave Alert\n");
    }
}

void Pop()
{
    int i;

    if (top == -1)
    {
        printf("No active alerts.\n");
    }
    else
    {
        printf("Removed Heatwave Alert: ");

        printf("%d", stack[top]);
        top--;
    }
}

void Display()
{
```

```
    int i;

    if (top == -1)
    {
        printf("No active alerts.\n");
    }
    else
    {
        printf("Active Heatwave Alerts:\n");

        for (i = top; i >= 0; i--)
        {
            if (stack[i] >= 45)
                printf("Extreme Red Heatwave Alert\n");
            else if (stack[i] >= 40)
                printf("Orange Heatwave Alert\n");
            else if (stack[i] >= 35)
                printf("Yellow Heatwave Alert\n");
        }
    }
}

int main()
{
    int choice;
    int temperature;

    do
    {
        printf("\n--- HEATWAVE ALERT SYSTEM ---\n");
        printf("1. PUSH Temperature\n");
        printf("2. POP Alert\n");
        printf("3. Display Alerts\n");
        printf("4. EXIT\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice)
        {
            case 1:
                printf("Enter temperature in Celsius: ");
                scanf("%d", &temperature);

                if (temperature >= 45)
                {
                    printf("Extreme Red Heatwave Alert!\n");
                    Push(temperature);
                }

                if (temperature >= 40 && temperature < 45)
                {
                    printf("Orange Heatwave Alert!\n");
                    Push(temperature);
                }
```

```
            if (temperature >= 35 && temperature < 40)
            {
                printf("Yellow Heatwave Alert!\n");
                Push(temperature);
            }

            if (temperature < 35)
            {
                printf("Temperature is Normal.\n");
                printf("Clearing all active alerts...\n");

                while (top != -1)
                {
                    Pop();
                }
            }
            break;

        case 2:
            Pop();
            break;

        case 3:
            Display();
            break;

        case 4:
            printf("Program Ended.\n");
            break;

        default:
            printf("Invalid Choice.\n");
    }

} while (choice != 4);

return 0;
}
```

Output :

```
C:\TURBOC3\BIN>TC

--- HEATWAVE ALERT SYSTEM ---
1. PUSH Temperature
2. POP Alert
3. Display Alerts
4. EXIT
Enter your choice: 1
Enter temperature in Celsius: 30
Temperature is Normal.
Clearing all active alerts...
```

```
--- HEATWAVE ALERT SYSTEM ---
1. PUSH Temperature
2. POP Alert
3. Display Alerts
4. EXIT
Enter your choice: 1
Enter temperature in Celsius: 40
Orange Heatwave Alert!
Orange Heatwave Alert
```

```
--- HEATWAVE ALERT SYSTEM ---
1. PUSH Temperature
2. POP Alert
3. Display Alerts
4. EXIT
Enter your choice: 2
Removed Heatwave Alert: 42
--- HEATWAVE ALERT SYSTEM ---
1. PUSH Temperature
2. POP Alert
3. Display Alerts
4. EXIT
Enter your choice: 3
Active Heatwave Alerts:
Orange Heatwave Alert

--- HEATWAVE ALERT SYSTEM ---
1. PUSH Temperature
2. POP Alert
3. Display Alerts
4. EXIT
```